

EVOLUTIONARY PRINCIPLES OF PLASTICITY REAPPEAR IN DEEP NEURAL NETWORKS

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ABSTRACT. Deep neural networks (DNNs) and genotype–phenotype (GP) maps both contain large *neutral sets*, or regions of parameter or genotype space that implement the same input–output behavior. In GP maps, the size and arrangement of these neutral sets strongly influence *plasticity* under fluctuating environments. Here we ask whether analogous geometric principles govern plasticity in DNNs trained by stochastic gradient descent. Using a fully connected network that realizes all 7-bit Boolean functions, we estimate an effective neutral set size (NSS) for each function using a simplicity-bias-based probability model. We then select pairs of functions spanning low, medium, and high values of both NSS and Hamming distance, and alternate training between each pair across a grid of learning rates and environmental switching periods. Plasticity is quantified through (i) the distance to a two-task “utopia” point, (ii) robustness to environmental switches, and (iii) the time required to reach a periodic steady state. A two-way ANOVA shows that both NSS and Hamming distance significantly affect all three measures, with Hamming distance exerting the dominant influence. Heatmaps reveal that decreasing Hamming distance and increasing NSS both enlarge the region of hyperparameters that yield robust, near-utopian plasticity. These results parallel recent findings in RNA GP maps, suggesting that neutral-set geometry provides a unifying framework for understanding adaptive flexibility in both evolutionary and gradient-based learning systems.

1. INTRODUCTION

This work explores the implications of three empirically supported hypotheses about the loss landscapes of deep neural networks (DNNs):

- (1) Local minima are approximately equivalent to one another, and to global minima, in terms of performance [CHM⁺14, LvB17].
- (2) Minima are connected by smooth paths along which the training and test loss remain essentially unchanged (“mode connectivity”) [GIP⁺18, YPS⁺22, LBD⁺23].
- (3) Minima that generalize well are surrounded by flat regions of the loss landscape [HS97, HVSG25, PRSA23].

We refer to the subset of parameter space that maps to these connected, similarly performing minima as the *neutral set* of model space for a given dataset.

This terminology is borrowed from evolutionary biology, and in particular from the study of genotype–phenotype (GP) maps. There, the neutral set denotes the family of genotypes that all give rise to the same phenotype (often a protein structure) [Ahn17]. Whereas DNN parameters are learned by gradient-based optimization, genotypes evolve under mutation and selection. Consequently, the number of mutations required to transform one genotype into another induces a natural geometry on the “parameter space” of GP maps via the *Hamming distance* d_H . For two binary genotypes p_1, p_2 of length L , this distance is

$$d_H(p_1, p_2) = \sum_{k=1}^L \delta(p_1^{(k)}, p_2^{(k)}) .$$

For fixed-length genotypes, the number of sequences that map to a given phenotype is called its *neutral set size* (NSS) and can be computed exactly [Ahn17]. The geometry induced by Hamming distances reveals that the neutral sets of many GP maps are highly connected, closely mirroring the mode connectivity phenomenon observed in DNNs. Unlike in the DNN setting, however, this connectivity can often be understood mechanistically through percolation theory [Ahn17]. Moreover, GP maps exhibit an evolutionary

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bias: within a neutral set, evolution tends to favor genotypes with higher robustness [GGA25a], indicating similarity to the low curvature around flat minima in DNNs.

These shared geometric features between desirable phenotypes in GP maps and high-performing DNN parameterizations motivate the hypothesis that further properties of GP maps may have analogues in deep learning. In particular, this study investigates whether the relationships between plasticity, neutral set size, and Hamming distance identified in [GGA25b] for evolving genotype populations extend to the parameter space of DNNs.

In particular, when GP-map evolution occurs under alternating environments, both the Hamming distance and the NSS of the phenotypes optimal in each environment strongly influence the emergence of *plasticity*: the ability of a population to reach genotypes that achieve high fitness in both environments. This work examines whether structurally similar dependencies arise when DNNs are alternately trained on two datasets.

2. METHODS

2.1. Measuring Neutral Sets. In the continuous parameter space of DNNs, calculating the NSS requires a different approach from the explicit combinatorial techniques used in GP maps. Instead of counting genotypes, we characterize the NSS through the probability density it occupies in parameter space. Although this density is typically intractable to compute directly, the problem simplifies when we restrict attention to n -bit Boolean functions

$$f : \{0, 1\}^n \rightarrow \{0, 1\}.$$

Following the technique of [MRVPL25], we estimate the probability mass of the neutral set associated with each Boolean function. For a parameter space Θ and a sampling distribution $P(\theta)$ over $\theta \in \Theta$, we threshold the real-valued outputs of a randomly initialized DNN to obtain a discrete function

$$F_\theta : \{0, 1\}^n \rightarrow \{0, 1\},$$

which must coincide with one of the 2^{2^n} possible Boolean functions. The NSS of a given function f is then the total probability mass of parameters whose thresholded network implements f :

$$(2.1) \quad P(f) = \int_{\{\theta | F_\theta = f\}} P(\theta) d\theta.$$

Prior work shows that for both DNNs [MRVPL25] and GP maps [JDG⁺22], this probability is strongly related to the Kolmogorov complexity of the output. Specifically, the following upper bound from algorithmic information theory holds [DCL18]:

$$(2.2) \quad P(f) \lesssim 2^{-\alpha \tilde{K}(f) + c},$$

where $\tilde{K}(f)$ denotes an empirical estimate of the Kolmogorov complexity of f . Systems that obey this exponential decay are said to exhibit *simplicity bias*: simpler outputs occupy disproportionately large volumes of parameter space.

In our setting, we approximate 7-bit Boolean functions using a DNN with 10 hidden layers of width 40 and hyperbolic tangent activations. Network parameters are sampled i.i.d. from a standard normal distribution. The complexity $\tilde{K}(f)$ of each Boolean function is estimated using a Lempel–Ziv (LZ) complexity measure following [DCL18]. The constants α and c in Equation 2.2 are obtained by fitting a 95% confidence upper envelope to the empirical scatter plot of $P(f)$ versus $\tilde{K}(f)$ (Fig. 1). We use the estimated upper bound as an empirical proxy for the NSS.

2.2. Plasticity. In fluctuating environments, the genotype–phenotype study of [GGA25b] examines how populations evolve when selection alternates between two environments (Y_1, Y_2) every ΔT generations. Using a genetic algorithm with mutation rate μ , the authors identify four key factors that govern the emergence of *plastic* genotypes capable of achieving high fitness in both environments:

- (1) the neutral set sizes (NSS) of the optimal phenotypes in each environment;
- (2) the Hamming distance d_H between genotypes within the corresponding neutral sets;
- (3) the environmental switching period ΔT ;
- (4) the mutation rate μ controlling exploration of genotype space.

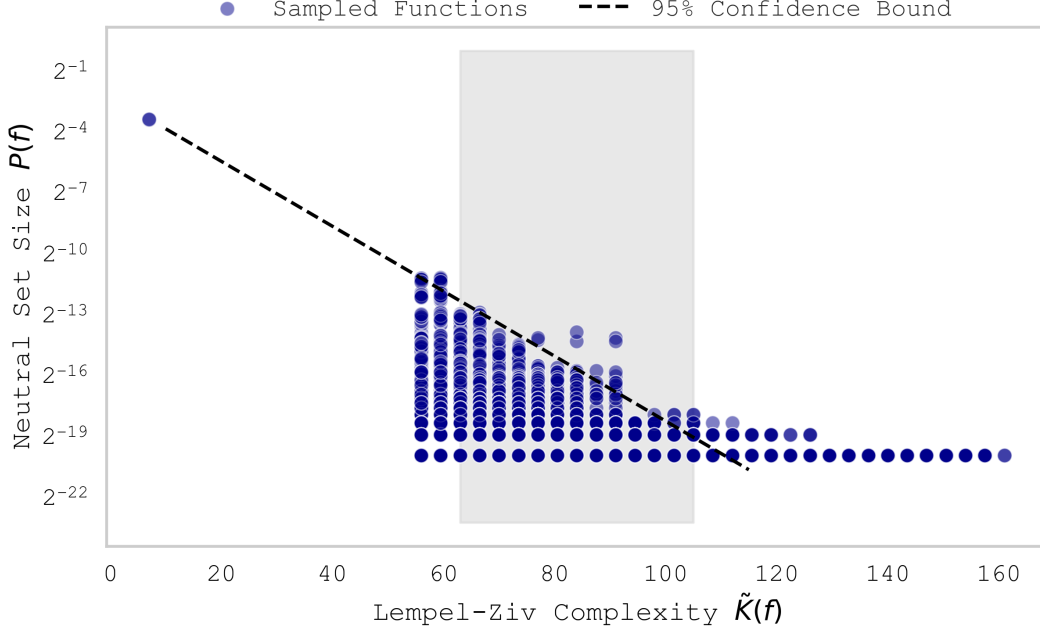


FIGURE 1. Bounding the NSS of a DNN approximating 7-bit Boolean functions. The 95% confidence upper envelope is given by $P(f) \leq 2^{-0.16\tilde{K}(f)+0.21}$. All Boolean functions used in subsequent experiments fall within the range of complexities indicated by the shaded region.

Plasticity in their framework is quantified both by the fraction of the population that attains high-fitness phenotypes in both environments, and by the distance of the population’s average performance to a two-dimensional utopia point corresponding to perfectly balanced fitness in both tasks.

A central finding of [GGA25b] is that plasticity is maximized when (i) the optimal phenotypes have *large* neutral sets, (ii) the Hamming distance between these neutral sets is *small*, and (iii) the switching period ΔT matches the characteristic exploration timescale determined by μ . This interaction creates a regime where the population repeatedly moves along well-connected neutral regions whose geometry supports rapid adaptation between environments.

Translating this picture from GP maps to DNNs requires reinterpreting each of the four factors. The “environments” become dataset labels, and “fitness” becomes performance with respect to the corresponding loss functions of each label set (here, binary cross-entropy). The NSS and Hamming distances are measured as described in the previous sections. The mutation rate μ of the genetic algorithm plays the role of the learning rate η in stochastic gradient descent, determining how aggressively parameters explore the loss landscape. Likewise, the environmental switching period ΔT corresponds to the number of SGD updates spent on each dataset before changing the labels used in the loss function.

Unlike genetic algorithms, SGD produces a single trajectory rather than a population, so the population fraction used in [GGA25b] does not have a natural analog. Instead, we focus on the geometric and dynamical properties of a *single parameter trajectory* as datasets alternate.

In the GP setting, the utopia distance measures how closely the population comes to performing equally well in both environments simultaneously. In our DNN setting, for a model with losses $\ell_1(t)$ and $\ell_2(t)$ at time t , we convert each loss into an accuracy proxy $a_i(t) = e^{-\ell_i(t)}$. The utopia point corresponds to $(1, 1)$, and the instantaneous utopia distance is

$$d_U(t) = \left\| \begin{pmatrix} a_1(t) \\ a_2(t) \end{pmatrix} - (1, 1) \right\|_2.$$

For each run, we report the *average utopia distance* over the final $2\Delta T$ steps of training:

$$d_U = \frac{1}{2\Delta T} \sum_{t=T-2\Delta T}^{T-1} d_U(t),$$

mirroring the time-averaged measure of plasticity used in [GGA25b].

To capture aspects of plasticity specific to gradient-based adaptation, we introduce two additional metrics. First, the *robustness* measures how dramatically accuracy changes when the environment switches. If $a_1(t)$ is the accuracy proxy on dataset 1, the robustness is

$$\Delta a_1 = \max_{t \in \text{last } 2\Delta T} a_1(t) - \min_{t \in \text{last } 2\Delta T} a_1(t),$$

Smaller robustness indicates that performance is stable across switches.

Second, we measure the length of training required for the alternating SGD trajectory to reach a stable periodic orbit under switching. In practice, this *steady state time* t_{SS} is defined as the earliest time after which consecutive periods differ by less than a tolerance.

These quantities jointly allow us to characterize plasticity analogously to [GGA25b]: the utopia distance reflects how well a single set of parameters performs on both tasks; the robustness captures the cost of switching environments; and the time to steady state captures how quickly the system adapts to recurring fluctuations.

2.3. Experimental Setup. To examine how neutral set size, Hamming distance, learning rate, and environmental switching period jointly influence plasticity, we construct a controlled set of alternating-training experiments on pairs of Boolean functions.

We begin by generating a library of 7-bit Boolean functions. For each function, we compute both its empirical neutral set size (via $\log_2 P(f)$ obtained from the simplicity-bias scaling law) and its Hamming distances to all other functions. From this collection we select a representative 3×3 grid of function pairs that span low, medium, and high values of both NSS and Hamming distance. The nine function pairs used in all experiments are shown in Table 1.

TABLE 1. Selected pairs of Boolean functions spanning low, medium, and high neutral set sizes and Hamming distances.

Pair	d_H	$\log_2(\text{NSS})$
(26, 27)	15.0	(−18.4915, −18.4915)
(24, 29)	69.0	(−17.9315, −19.0515)
(22, 28)	95.0	(−17.3715, −19.0515)
(12, 13)	2.0	(−14.5715, −14.5715)
(9, 19)	70.0	(−14.0115, −16.2515)
(7, 8)	121.0	(−14.0115, −14.0115)
(0, 1)	3.0	(−11.7715, −11.7715)
(1, 3)	66.0	(−11.7715, −12.3315)
(6, 8)	121.0	(−13.4515, −14.0115)

For each Boolean function f , the full set of 2^7 inputs is used as the dataset, with $X \in \{0, 1\}^{128 \times 7}$ and binary labels $Y_f \in \{0, 1\}^{128}$. All experiments use the same fully connected neural network architecture as in the NSS estimation stage. Parameters are initialized from the Gaussian prior described in [MRVPL25], with variance scaled by layer width.

Each experiment alternates SGD training between two functions and their labels Y_f in Table 1. Training begins with a pretraining phase on the first function until the loss reaches a tolerance or a preset iteration limit. After pretraining, the environment alternates between the two functions every ΔT SGD steps until either a steady state or the maximum 2500 iterations are reached. At each step, the losses on both environments are recorded and converted to accuracy proxies $a_i(t) = e^{-\ell_i(t)}$.

To explore the interaction between exploration scale and switching timescale, we sweep over six learning rates

$$\eta \in \{10^{-4}, 10^{-3.5}, 10^{-3}, 10^{-2.5}, 10^{-2}, 10^{-1.5}\},$$

and six switching periods

$$\Delta T \in \{1, 5, 10, 25, 50, 100\}.$$

For each of the nine function pairs and each $(\eta, \Delta T)$ combination, we run ten Monte Carlo trials with independent random initializations, for a total of $9 \times 6 \times 6 \times 10 = 3240$ trajectories.

For every trajectory we compute the three plasticity-related quantities (utopia distance d_U , robustness Δa_1 , time to steady state t_{SS}). These measurements provide a comprehensive dataset relating plasticity to both geometric and dynamical properties of learning. An example trajectory with the three plasticity measures is presented in Fig. 2.

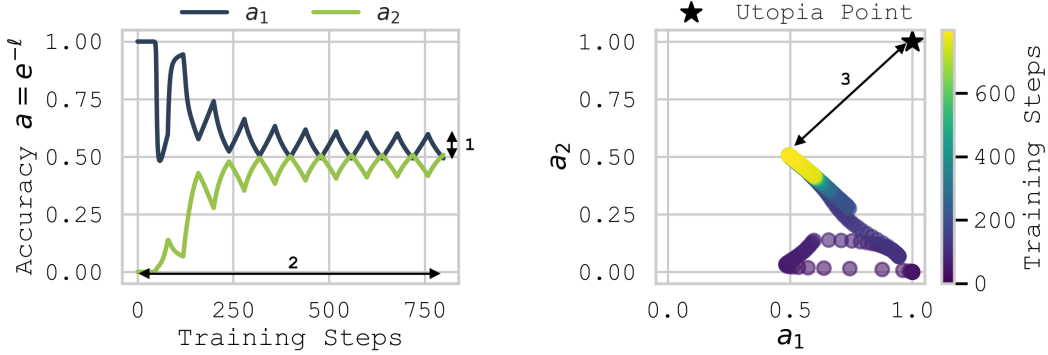


FIGURE 2. Example trajectory for $\eta = 0.001$, $\Delta T = 40$ using pair (9,19) from Table 1. The annotated arrows 1, 2, and 3 represent the robustness Δa_1 , steady state time t_{SS} , and utopia distance d_U respectively.

3. RESULTS

First, a two-way ANOVA was run to test the effect of NSS and Hamming distance on the three metrics of plasticity (Table 3). Across all metrics, the impacts of both NSS and Hamming distance were significant. The Hamming distance d_H emerged as the dominant explanatory factor, yielding extremely large F-statistics and vanishingly small p -values. For utopia distance d_U , both main effects were highly significant, and the interaction term was also strong, indicating that the influence of NSS depends on the mutational distance between functions. Robustness Δa showed a similar pattern: d_H had a large effect, NSS contributed a smaller but significant effect, and their interaction remained significant, suggesting nonlinear coupling between proximity in function space and neutral-set density. For steady-state time t_{SS} , d_H was again highly significant, NSS retained a modest but meaningful effect, and the interaction was not statistically significant at the $\alpha = 0.05$ level. Together, the ANOVA confirms that the geometry of the function space (captured by Hamming distance and NSS) shapes all three forms of plasticity, with d_H exerting the strongest overall influence.

Next, to examine how these structural factors interact with the training dynamics $(\eta, \Delta T)$, heatmaps were constructed for utopia distance d_U and robustness Δa (Fig. 3). Across all nine (d_H, NSS) pairs, both metrics reveal systematic and interpretable patterns. For both metrics, low values indicate higher plasticity.

For robustness Δa , a consistent trend emerges: small learning rates and large switching periods (upper left of each panel) produce the most robust plastic behavior (low Δa). As Hamming distance decreases, this region of high robustness expands, making plasticity accessible over a wider range of $(\eta, \Delta T)$. Increasing NSS produces a similar but weaker effect, slightly enlarging the basin of hyperparameters that yield robust switching.

For utopia distance d_U , the strongest effects appear at low Hamming distance. When the two target functions differ by only a few bits, large NSS dramatically lowers d_U across nearly the entire $(\eta, \Delta T)$ grid, meaning that plastic solutions close to the utopia point become easily attainable. At medium Hamming distance, this effect persists but is noticeably dampened. At high Hamming distance, the dependence on NSS nearly disappears and the achievable d_U varies primarily with $(\eta, \Delta T)$.

TABLE 2. ANOVA results for the effects of Hamming distance d_H and neutral set size (NSS) on utopia distance d_U , robustness Δa , and steady-state time t_{SS} . Reported are the F-statistics, and p -values.

Metric	Effect	F	p
d_U	d_H	7730.22	$< 10^{-300}$
	NSS	593.40	3.19×10^{-220}
	$d_H \times \text{NSS}$	356.16	2.56×10^{-254}
Δa	d_H	388.60	5.90×10^{-152}
	NSS	5.58	3.79×10^{-3}
	$d_H \times \text{NSS}$	15.15	2.80×10^{-12}
t_{SS}	d_H	89.47	1.51×10^{-38}
	NSS	5.25	5.30×10^{-3}
	$d_H \times \text{NSS}$	2.04	8.61×10^{-2}

Overall, decreasing the Hamming distance between environments (bringing the two target functions closer together) has an effect analogous to, but substantially stronger than, increasing NSS. Both factors shift the system toward enhanced plasticity, but Hamming distance reshapes the entire landscape of attainable $(\eta, \Delta T)$ regimes, making plasticity achievable at faster learning rates and slower environmental switching. In contrast, NSS mainly amplifies plasticity when the two environments are already structurally similar.

4. DISCUSSION

4.1. Comparison with Genotype–Phenotype Plasticity. Our results demonstrate that the geometry of the Boolean-function landscape exerts a strong and systematic influence on plasticity in deep neural networks. Across all analyses, Hamming distance d_H between functions emerged as the dominant explanatory factor:

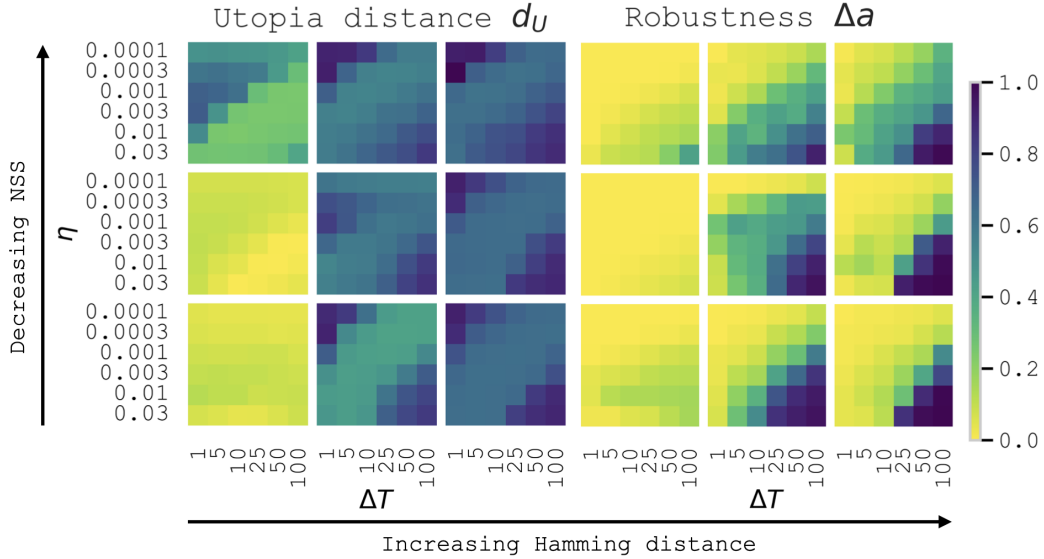


FIGURE 3. Heatmaps of plasticity measures across learning rates η and environmental switching periods ΔT for all selected (d_H, NSS) pairs. Left block: utopia distance d_U , measuring joint performance across both datasets. Right block: robustness Δa , quantifying the accuracy drop immediately after an environmental switch. Each pixel shows the mean over 10 Monte Carlo runs.

smaller d_H consistently reduced utopia distance d_U , improved robustness Δa , and shortened steady-state time t_{SS} . Neutral set size (NSS) played a secondary but still meaningful role, particularly in shaping the region of $(\eta, \Delta T)$ space in which plasticity was accessible. Larger NSS widened this region and generally flattened the response surface over learning rates and switching periods. The two factors also exhibited significant interactions for both d_U and Δa , indicating that the influence of NSS is magnified when functions lie close together in Hamming space. Together, these findings show that the emergence of plasticity in DNNs is governed by the same two geometric features that organize genotype landscapes: the size of neutral sets and the mutational (or functional) distances between them.

These patterns parallel the findings from the GP-map study of Ahnert and coauthors [GGA25b]. They find that plasticity arises most readily when the neutral networks corresponding to the two environments have (i) large neutral set sizes and (ii) short mutational distances between them. Large NSS produces extended, percolating neutral networks that allow populations to diffuse widely without loss of fitness, while small Hamming separations ensure that movement between the two phenotypic optima does not require traversing deep fitness valleys. The GP-map study further showed that when these geometric conditions are met, short environmental switching periods ΔT promote plasticity by preventing fixation at a single phenotypic optimum, and that the mutation rate μ interacts with ΔT to determine whether populations track both environments or specialize in one.

Our DNN results reproduce these qualitative dependencies despite the radically different underlying mechanisms. In place of mutation and selection, parameter updates are governed by stochastic gradient descent; in place of discrete genotypes, we consider continuous parameter vectors with a Boolean thresholding map; and in place of population diffusion, the system evolves along a single SGD trajectory. Yet the macro-scale phenomena persist. When the two datasets correspond to functions of low Hamming distance, SGD readily finds trajectories that reach high-performing states in both environments and transitions between them occur with little loss. This mirrors the rapid genotype switching seen in GP maps when two phenotypes are separated by only a few mutations. Similarly, increasing NSS in our setting expands the region of plastic behavior by creating broader basins of parameters that implement each Boolean function, closely analogous to the way extended neutral networks in GP maps buffer the effects of environmental fluctuations.

There are, however, notable divergences that arise from the differences between evolutionary dynamics and SGD. In GP maps, large NSS not only facilitates plasticity but also increases evolutionary robustness by reducing the likelihood that random mutations damage the phenotype. In contrast, in the DNN setting, large NSS primarily modulates the accessibility of plastic trajectories rather than conferring robustness in a literal mutational sense. Moreover, the effect of NSS in our experiments is strongly dependent on d_H : when the Hamming distance is large, NSS has little measurable effect on either utopia distance or robustness, whereas in GP maps the two factors contribute more symmetrically to plasticity. This suggests that the functional geometry of Boolean neural approximators is more sharply constrained than classical genotype spaces, likely reflecting the high-dimensional but highly structured nature of neural parameterizations.

Despite these differences, the central conclusion aligns across both systems: *plasticity is fundamentally a geometric consequence of how neutral sets are arranged in the space of possible functions*. When neutral sets are broad and closely positioned, plastic behavior emerges naturally under both SGD and evolutionary dynamics. When they are small or widely separated, plasticity becomes fragile or unattainable. The remarkable agreement between these models, one biological, one computational, highlights the possibility that neutral-set geometry provides a unifying framework for understanding adaptive flexibility across learning systems, whether implemented through genetic mutation or gradient descent.

4.2. Implications for Learning Systems. The plasticity experiments above are defined in terms of two environments, but the same geometry naturally appears in standard supervised learning. A training set and a test set can be viewed as two environments sampled from related input-output maps. When the underlying functions are close in an appropriate distance, a single network can achieve low loss on both, while large discrepancies make this impossible regardless of optimization details. In this view, generalization is a special case of plasticity where the two environments correspond to empirical and true risk, and the relevant distance is small by construction.

Our results suggest that in this regime of small functional distance, neutral set structure becomes especially important. When the Hamming distance between the two Boolean functions is low, increasing neutral set

size sharply improves utopia distance and robustness, whereas at large Hamming distance NSS has almost no effect. This is consistent with the simplicity bias observed for DNNs and other input–output maps, where simple outputs with large pre-images under the parameter map occur with exponentially higher probability than complex ones [DCL18, MRVPL25, JDG⁺22]. In the GP-map setting, large neutral networks support both robustness and evolvability in fluctuating environments [Ahn17, GGA25b]. Translating to learning theory, functions with large effective NSS should be easier to fit on training data and more likely to generalize when the train–test distance is small, because many nearby parameterizations implement similar hypotheses that perform well on both environments. Further, the link between NSS and complexity [DCL18] provides a heuristic way of estimating NSS for more complicated scenarios than the simplified Boolean case presented here.

Plasticity also provides a geometric link between generalization and the loss landscape. In the RNA genotype–phenotype map, phenotypic cliffs separate large regions of high fitness from sharp drops in performance, and robust phenotypes occupy extended plateaus buffered from these cliffs [GGA25a]. Analogous structure is now well documented in DNN loss landscapes, where flat minima, low curvature regions, and connected basins correlate with good test performance [HS97, KMN⁺17, PRSA23, GIP⁺18, YPS⁺22, LBD⁺23, LG25, HVSG25]. In our experiments, robustness $\Delta\alpha$ measures how much performance degrades when the environment changes, which is closely related to the sensitivity of the loss under perturbations. High robustness therefore implies a locally flat joint loss surface over both environments, while low robustness indicates that the trajectory lies near a plasticity cliff where small changes in data distribution lead to large changes in performance. This is directly analogous to the role of cliffs and neutral plateaus in GP maps, and is naturally connected to information-geometric notions such as Fisher information and local curvature [Nie18, JNM⁺20].

Taken together, these observations suggest a unified picture in which neutral-set geometry, simplicity bias, and loss-landscape flatness are different faces of the same underlying structure. For real-world learning problems that go beyond Boolean functions, one can view generalization as a form of plasticity between closely related environments, with NSS-like measures controlling how large a set of parameters achieve low joint loss. This perspective points toward new generalization diagnostics that explicitly combine functional distance between environments with neutral set density and curvature-based measures of robustness, and it offers a route for importing tools from genotype–phenotype theory into the analysis and design of deep learning systems.

5. CONCLUSION

This study demonstrates that plasticity in deep neural networks is shaped by geometric properties that closely parallel those governing plasticity in genotype–phenotype maps. Despite differences between evolutionary dynamics and stochastic gradient descent, the emergence of plasticity in both systems appears to be governed by the same underlying geometry of neutral sets. This suggests that plasticity and generalization in DNNs may be understood through a unified framework that blends simplicity bias, neutral-set density, and loss-landscape (fitness landscape) flatness. In particular, when environments (or datasets) are functionally similar, large neutral regions support robust, stable performance across changes, providing a geometric lens on why networks trained on simple or structured tasks tend to generalize well.

Future work should extend these ideas beyond Boolean functions to richer datasets and architectures, develop practical proxies for NSS in large-scale models, and explore how modifying architectures or priors might explicitly shape neutral-set geometry. Such directions may offer new tools for designing learning systems that are inherently more robust, adaptive, and capable of transferring knowledge across changing tasks.

Data and Code Availability. The implementation and scripts to reproduce all experiments are available at https://github.com/oskril01/DNN_Plasticity.

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